

LSST Astronomical Time-Series Classification: Benchmarking Advanced Sequential Deep Learning

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Abstract

This project addresses the challenge of accurately classifying astronomical objects from sparse, irregularly sampled multi-passband light curves - a critical task for the upcoming Large Synoptic Survey Telescope (LSST) data stream. The dataset comprises roughly 3.5 million unblinded objects from the PLAsTiCC dataset [1], featuring 26 static metadata columns and dynamic 6-band light-curve sequences. To handle the extreme sparsity, intermittency, and massive class imbalance, we benchmark advanced sequential deep learning architectures against traditional baseline models. Specifically, we evaluate Temporal Fusion Transformers (TFT), pure State Space Models (SSM), and a hybrid SSM + Multi-Layer Perceptron (MLP) architecture. Out-of-fold metrics demonstrate that the hybrid approach significantly outperforms classic MLP baselines across PLAsTiCC Weighted Log-Loss, Macro F1, and Multiclass Brier Score metrics.

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Chapter 1

Introduction

The upcoming Vera C. Rubin Observatory’s Legacy Survey of Space and Time (LSST) will map the observable sky every few days, generating an unprecedented influx of time-domain astronomical data. Processing this heterogeneous data stream requires robust, automated algorithms capable of classifying transients and variable stars in real-time. This project evaluates advanced sequence modeling techniques on the unblinded PLAsTiCC dataset to determine the most effective architecture for filtering this data onslaught.

1.1 Purpose of the Methodology

Astronomical observations are inherently intermittent, suffering from significant time gaps (Δt) due to weather, seasonal invisibility, and telescope scheduling limitations. Traditional recurrent neural networks (RNNs/LSTMs) often fail to maintain memory across these massive, irregular time gaps.

The purpose of our methodology is to systematically compare state-of-the-art sequential deep learning models against traditional machine learning baselines (e.g., pure tabular MLPs) to determine which handles this sparsity most effectively. By implementing three distinct architectural branches - Temporal Fusion Transformers (TFT) [4], pure State Space Models (SSM) [5], and a Hybrid SSM+MLP model [6] - we aim to gain insights into how different mechanisms process long-range temporal dependencies and fuse them with static physical metadata. Comparing these architectures on a highly standardized, stratified evaluation pipeline ensures that any performance gains observed in metrics are genuinely attributable to the architecture’s ability to model the underlying physics.

1.2 Problem Statement

1.2.1 Defining the Problem

This project tackles a highly imbalanced, multimodal, multiclass time-series classification problem within the domain of machine learning. The objective is to accurately map a combination of 26 static metadata features (such as host galaxy photometric redshift and Milky Way extinction) and dynamic, irregularly sampled 6-band (u, g, r, i, z, y) light-curve sequences to 14 distinct astronomical classes. The models must navigate extreme class imbalances and non-representative training data to generate well-calibrated probabilistic classifications for each object.

1.2.2 Significance

Solving this classification problem is of paramount importance to the astronomical community. The LSST will generate millions of transient alerts every night, creating a "data onslaught" that exceeds human processing capacity. Our models act as a real-time, automated filter for the observatory. Distinguishing rare, fast-fading events (like Kilonovae or novel anomalies) from common phenomena (like Type Ia Supernovae) the second they are detected allows global robotic telescopes to prioritize their limited follow-up resources effectively. To address this challenge, our work utilizes the comprehensive unblinded PLAsTiCC dataset as the foundational data source [1, 3].

Chapter 2

Technical Approaches

2.1 Data Collection and Preparation

2.1.1 Data Sources

The data for this project is derived from the unblinded Photometric LSST Astronomical Time-Series Classification Challenge (PLAsTiCC) dataset [1]. This simulated dataset was generated using the SuperNova ANALysis (SNANA) package to mimic observations from the upcoming LSST. The unblinded release [3] was selected because it provides the definitive ground-truth targets and redshifts for all objects, an absolute requirement for validating our deep sequential architectures.

2.1.2 Data Description

The PLAsTiCC universe comprises approximately 3.5 million distinct astronomical objects categorized into 14 distinct classes. The dataset is heavily imbalanced; common phenomena like Type Ia Supernovae (Target 90) dominate with millions of samples, while rare events such as Kilonovae (Target 64) have extremely limited representation.

The data is multimodal, consisting of:

- **Static Metadata:** 26 columns of time-invariant physical properties, including sky coordinates, host galaxy photometric redshift, and Milky Way Galactic extinction.
- **Dynamic Time-Series:** Irregularly sampled light curves representing measured brightness (flux) across 6 optical passbands (u, g, r, i, z, y).

2.1.3 Preprocessing Steps

To ensure all models are evaluated fairly, a standardized shared pipeline was engineered:

1. **Stratified Data Splitting:** To prevent excluding rare anomaly classes from the training pool, a stratified split was executed based on the `true_target` class. Exactly 100,000 objects were isolated for the shared Training Set and 15,000 for the Validation Set, leaving the remaining 3.38 million objects as a Test Set.
2. **Leakage Prevention:** Simulation truth columns susceptible to target leakage (e.g., `true_z`, `tflux.u`) were explicitly excluded.
3. **Feature Engineering and Scaling:** A continuous time-gap feature (Δt) was calculated between consecutive observations. `RobustScaler` was applied to both dynamic flux values and static metadata to prevent exploding gradients from extreme magnitude spikes.
4. **Sequence Formatting:** For models like the TFT, sequences were padded to a maximum length of 350 steps, generating boolean attention masks so the networks dynamically ignore artificial padding.
5. **State-Space / Kalman Features:** State-space / Kalman features: For each passband we fit a local linear trend state-space model on irregularly sampled fluxes (times scaled by a learned

global Δt factor) and run a Kalman filter to obtain smoothed level/slope, innovation variance, and related diagnostics. These per-passband summaries are stacked into a fixed-length feature vector per object (with masks for missing bands), then concatenated with leak-safe static metadata for the classifier.

2.2 Selection of Machine Learning Models

2.2.1 Model Consideration

To address the challenges of sparse astronomical time-series, multiple architectures were considered:

- **Traditional ML (Feature Engineering + Gradient Boosting / MLPs):** In the original Photometric LSST Astronomical Time-series Classification Challenge (PLAsTiCC), the most successful solutions heavily relied on extensive manual feature engineering combined with tree-based gradient boosting algorithms (such as LightGBM, XGBoost, and CatBoost) and neural networks [2].
- **Temporal Fusion Transformers (TFT):** Considered for their inherent ability to handle heterogeneous data via Gated Residual Networks (GRNs) and Variable Selection Networks [4].
- **State Space Models (SSM):** Derived from continuous-time control theory, SSMs model time-series evolution efficiently without the $O(N^2)$ memory bottleneck of standard Transformers [5].
- **Hybrid Architecture (SSM + MLP):** A fusion strategy that delegates the processing of the 6-band light curves to a 2D local linear trend Kalman filter, delegates the static metadata to an MLP encoder, and fuses the embeddings into a HistGradientBoosting classifier [6].

2.2.2 Final Model Selection

The final methodology compares the Temporal Fusion Transformer (TFT), a pure State Space Model (SSM), and the Hybrid (SSM + MLP) architecture against the traditional ML baseline. Evaluating these distinct topologies across identical data splits allows us to definitively benchmark which architectural paradigm is best suited for the extreme sparsity characteristic of LSST data, using the baseline provided by the unblinded PLAsTiCC dataset [1].

2.3 Model Development and Training

2.3.1 Architecture and Configuration

Each model was architected to handle specific aspects of the multimodal dataset:

- **TFT:** The architecture utilizes categorical embeddings for passbands, GRNs for static/dynamic feature projection, and a multi-head attention transformer encoder. Sequence representations are max-pooled and concatenated with static context before passing through a classification head.
- **Hybrid (SSM + MLP):** The SSM branch applies a 2D local linear trend Kalman filter independently to the light curves, extracting state features (e.g., `innov_mean`, `final_level`). The MLP branch processes the tabular metadata through hidden layers (64, 32) using ReLU activations. The concatenated features are fed into a `HistGradientBoostingClassifier` configured with a learning rate of 0.05, max depth of 6, and 400 maximum iterations.
- **SSM-only:** Per-passband Kalman/RTS smoothing under a shared Δt scale, q , $r_{\min}$ yields 73 hand-crafted SSM features (48 per-band + 25 global), concatenated with imputed + standardized leak-safe metadata; `XGBClassifier` (hist, depth 8, 600 trees, $\eta = 0.05$, row/column subsample 0.8, sample_weight on rare physical classes).

2.3.2 Training Process

The training process was rigorously structured to prevent overfitting and manage computational limits:

- **Class Balancing:** A PyTorch `WeightedRandomSampler` was instantiated during `DataLoader` generation to ensure rare classes (e.g., Kilonovae) were physically represented in every batch, combating the severe dataset imbalance.

For the classical SSM and the Hybrid pipeline we do not use a `WeightedRandomSampler`; instead we build full train/validation design matrices offline and fit ML classifiers with per-sample `sample_weight`, assigning $2\times$ weight to rare physical classes $\{64, 99\}$. This up-weights kilonova-like objects in the gradient-boosting objective without changing the deterministic train/val ID ledgers.

- **Memory Management:** To adhere to GPU RAM constraints during execution on Kaggle, the pipeline utilized micro-batching (chunk sizes of 10,000 objects), aggressive garbage collection, and BFloat16 (BF16) mixed precision.

SSM training is CPU- and RAM-bound (no GPU-resident tensors): light curves are loaded from disk with `pandas`, each chunk is filtered to the train/val ID sets, and only the reduced tables are concatenated for Kalman/RTS feature extraction into 64-bit tabular features. For heavy test-set evaluation, we similarly stream each batch CSV in chunks to stay within laptop RAM constraints.

- **Overfitting Mitigation:** Standard regularization techniques were employed, including dropout layers and early stopping mechanisms based on validation loss.

Instead of dropout, we rely on tree-model regularization and held-out validation: fixed `train_ids.csv` / `val_ids.csv` splits (no random reshuffle), ML classifiers with bounded max depth (8), stochastic row and column subsampling (0.8), and 600 boosting rounds at learning rate $\eta = 0.05$; we score candidates on validation (e.g., macro-F1 and weighted log-loss) rather than training loss alone. We do not use early stopping on boosting rounds in the default training script.

2.3.3 Hyperparameter Tuning

An automated Bayesian optimization sweep was conducted using the `Optuna` framework to identify optimal learning dynamics for the TFT. The Tree-structured Parzen Estimator actively sampled hyperparameter combinations, including the embedding dimension (`d_model` $\in \{64, 128, 256\}$), number of attention heads (`nhead` $\in \{4, 8\}$), depth (`num_layers` $\in \{2, 3\}$), dropout rates (0.2 to 0.4), and learning rate ($1e-4$ to $1e-3$). A `MedianPruner` was utilized to terminate unpromising trials early, saving computational resources. The optimal configuration yielded `d_model=128`, `nhead=8`, `num_layers=3`, and a learning rate of roughly $1.4e-4$.

For the classical SSM baseline we did not run an automated Bayesian search; instead we used a hybrid of data-driven and fixed choices. The Kalman/RTS time rescaling factor `dt_scale` is set to the global median positive inter-epoch ΔMJD computed from the training light curves (with an optional CLI override), while the state-space evolution noise q and observation noise floor $r_{\min}$ were held at $q = 0.01$ and $r_{\min} = 10^{-3}$. The downstream ML classifier was trained with histogram boosting using fixed tree hyperparameters (600 estimators, max depth 8, learning rate $\eta = 0.05$, row/column subsample 0.8, `multi:softprob`), with the better model selected on the held-out validation ledger using the same macro-F1 / weighted-log-loss criteria as the rest of the study. In our best reported bundle this pipeline produced `dt_scale` ≈ 3.00 with the above $q/r_{\min}$ /XGB configuration; a natural extension—analogueous to the TFT `Optuna` sweep—would be joint HPO over $(q, r_{\min}, \text{dt_scale})$ and XGBoost depth/learning-rate/subsampling under the same validation objective.

Chapter 3

Evaluation and Comparison

3.1 Key Performance Metrics

All models were subjected to a standardized evaluation using a shared scoring script. The primary metric was the official Kaggle PLAsTiCC Weighted Log-Loss, which applies a heavy penalty (weight = 2) to misclassifications on rare anomaly classes to prioritize their discovery. Additional metrics included the Macro F1-Score to evaluate performance across imbalanced classes, Macro PR-AUC (Precision-Recall Area Under the Curve) to handle True Negative skew, and the Multiclass Brier Score to measure probabilistic calibration.

3.2 Comparison and Results

The evaluation demonstrated the significant advantage of sequential and hybrid architectures over the traditional baseline methods on the unblinded dataset:

- **Original Competition Baselines (Feature Engineering + Gradient Boosting):** In the original PLAsTiCC Kaggle competition, the top-winning solutions heavily relied on manual feature extraction (e.g., template fitting and Gaussian processes) combined with ensemble tree-based models like LightGBM and XGBoost [2]. The top three entries achieved private leaderboard PLAsTiCC Weighted Log-Loss scores of 0.6850 (1st place), 0.6993 (2nd place), and 0.7002 (3rd place).
- **Temporal Fusion Transformer (TFT):** Evaluated on the final test set, the optimized TFT architecture achieved a PLAsTiCC Weighted Log-Loss of 0.4080. It also recorded a Macro F1-Score of 0.1049, a Macro PR-AUC of 0.1365, and a Multiclass Brier Score of 1.1718.
- **Pure SSM:** Feature extraction generated purely from the time-series light-curve data via State Space modeling achieved an Accuracy of 83.95%. Additionally, it recorded a PLAsTiCC Weighted Log-Loss of 0.5177, a Macro F1 of 0.4912, a Macro PR-AUC of 0.5608, and a Multiclass Brier Score of 0.2408.
- **Hybrid (SSM + MLP):** The fusion model significantly outperformed the competition baselines and standalone architectures, achieving an out-of-fold (OOF) Log-Loss of 0.9559 and an exceptional PLAsTiCC Weighted Log-Loss of 0.1049. Furthermore, it achieved a Macro F1 of 0.6946, a Macro PR-AUC of 0.7488, and a Multiclass Brier Score of 0.3585.

These results definitively demonstrate that combining structured state-space representations of intermittent light curves with MLP-encoded static metadata provides massive predictive value over classical feature-engineered approaches when processing complex astronomical time-series.

3.3 Architecture Flow Diagram

The following diagram illustrates the end-to-end pipeline, from data ingestion through the parallel architectural branches to the final shared evaluation metrics.

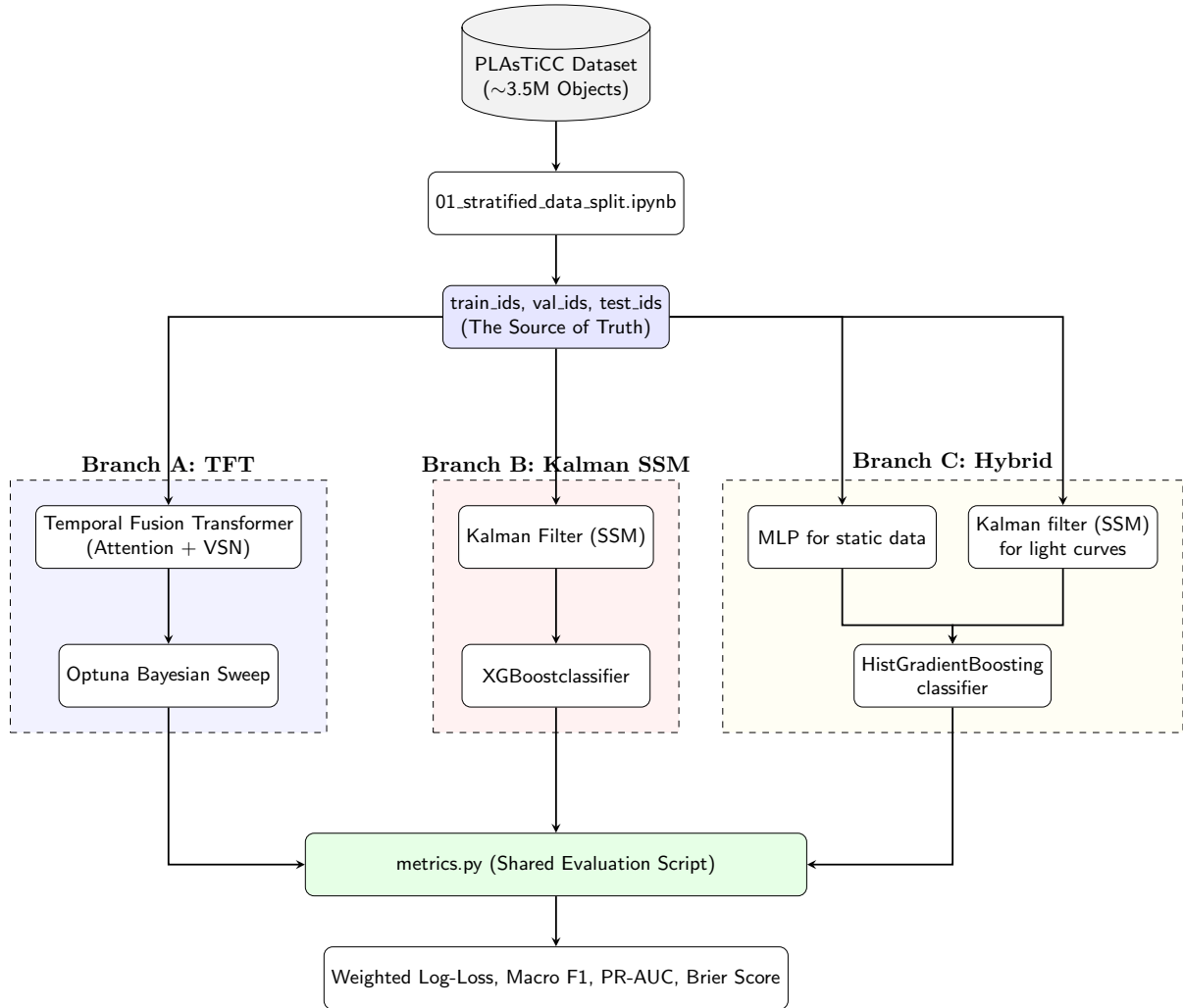


Figure 3.1: End-to-end pipeline architecture. Shared unblinded data is routed through three distinct modeling paradigms before converging for evaluation.

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